Stabilizing quantities ρ_p for boundary control of PDE

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Context

This is a question originally raised by Jean-Michel Coron in 2016. When studying the stability of nonlinear PDE of the form

$$\partial_t u + A(u)\,\partial_x u + B(u) = 0, \quad \text{on } [0, L], \tag{1}$$

where $u \in \mathbb{R}^n$ and with boundary conditions of the form

$$\begin{pmatrix} u_+(t,0) \\ u_-(t,L) \end{pmatrix} = G \begin{pmatrix} u_+(t,L) \\ u_-(t,0) \end{pmatrix}, \tag{2}$$

a natural quantity that appears is the following function

$$\rho_p : M \mapsto \inf\big\{\|\Delta M \Delta^{-1}\|_p \;\big|\; \Delta = \mathrm{diag}(\delta_1, \ldots, \delta_n),\ \delta_i > 0,\ \forall\, i \in \{1, \ldots, n\}\big\}, \tag{3}$$

where $\|\cdot\|_p$ is the matrix p-norm.

Conjectures

Conjecture 1. *For any $M \in \mathbb{R}^{n\times n}$ one has*

$$\rho_2(M) \le \rho_3(M) \le \cdots \le \rho_\infty(M).$$

A weaker version is:

Conjecture 2. *For any $M \in \mathbb{R}^{n\times n}$ one has*

$$\rho_2(M) \le \rho_k(M), \quad \forall\, k \in \mathbb{N} \setminus \{0\}.$$

Conjecture 3. *For any $n > 6$ and any $p > 2$ ($n, p \in \mathbb{N}$), there exists $M \in \mathbb{R}^{n\times n}$ such that*

$$\rho_p(M) < \rho_{p+1}(M). \tag{4}$$

What is already known

The main previous works are in [2, 1]. In particular it is known that for any $n \in \mathbb{N}$ and any $M \in \mathbb{R}^{n\times n}$,

$$\rho_1(M) \geq \rho_2(M), \quad \rho_1(M) = \rho_\infty(M),$$

and for $n \geq 2$ there exists $M \in \mathbb{R}^{n\times n}$ such that

$$\rho_1(M) > \rho_2(M).$$

It is also known [3] that if M has only non-negative coefficients then

$$\rho_k(M) = \rho_1(M), \quad \forall\, k \in [1, +\infty].$$

References

[1] Jean-Michel Coron and Hoai-Minh Nguyen. Dissipative boundary conditions for nonlinear 1-D hyperbolic systems: sharp conditions through an approach via time-delay systems. *SIAM J. Math. Anal.*, 47(3):2220–2240, 2015.

[2] Jean-Michel Coron, Georges Bastin, and Brigitte d'Andréa-Novel. Dissipative boundary conditions for one-dimensional nonlinear hyperbolic systems. *SIAM Journal on Control and Optimization*, 47(3):1460–1498, 2008.

[3] Siegfried M. Rump. Optimal scaling for p-norms and componentwise distance to singularity. *IMA Journal of Numerical Analysis*, 23(1):1–9, 2003.